ML in Investment Management

Dr. Eric Tham
Email: eric.tham@ntu.edu.sg

Outline

- Portfolio Investment
  - Risks and Returns in Finance
  - CAPM in Finance
  - Factor Pricing theory
- AI/ML methods applied
  - Logistic Regression (Binomial/ Multinomial)
  - Boosting Methods

Which Investment would You Choose?

Both investments cost $100 now

- Investment A: 100% chance of getting $150 in one year's time
- Investment B: 50% chance of getting $300 and 50% of getting $0 in one year's time

Investors' Attitude towards Risk

- Most investors are risk averse
  Risk-averse investors dislike risk and require higher rates of return to encourage them to hold riskier securities.
- Risk premium—the difference between the return on a risky asset and a riskless asset
  Serves as compensation for investors to hold riskier securities.

Does not mean that they will always choose the lowest risk investment!

Implication:
Higher risk ➔ Higher required return

Investment Returns

Calculating the rate of return on an investment:

$$\text{Return} = \frac{(\text{Ending value} - \text{Amount Invested})}{\text{Amount Invested}}$$

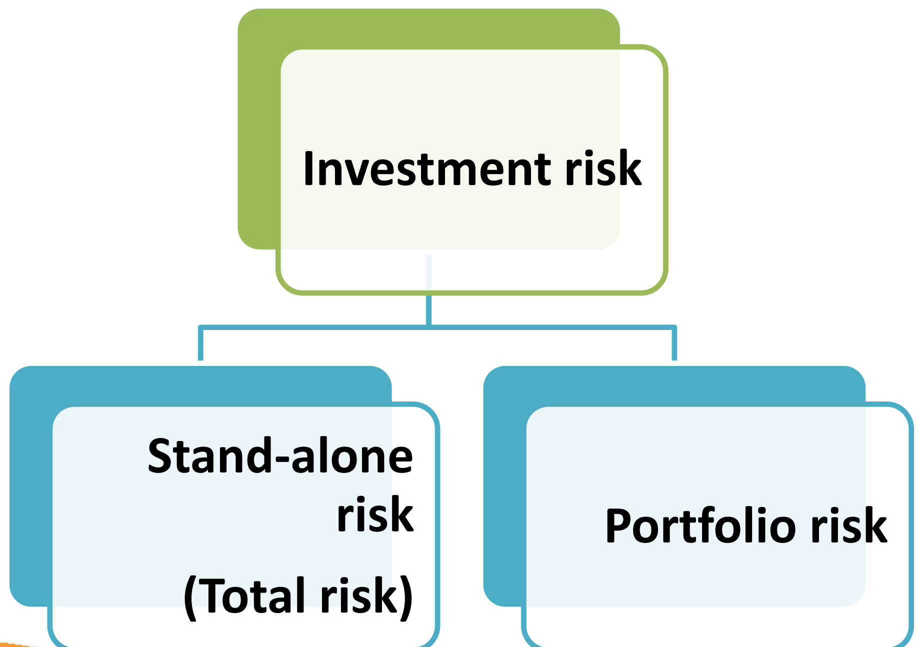
If $1,000 is invested and $1,100 is returned after one year, the annual rate of return is:

(1100–1000)/1000=10%

A stock is selling for $30 and you expect the stock price to be $39 in one year's time. The stock is also expected to pay a dividend of $3 at the end of the year. The annual rate of return is: (39+3-30)/30 = 40%

What is Investment Risk?

- Investment risk is related to the probability of earning a return that is different from expected.
  - Comprises both a single stock and all the stocks in a portfolio.
  - Measured by the <u>standard deviation</u> of the returns



Investment Alternatives

Market Index represents the general market

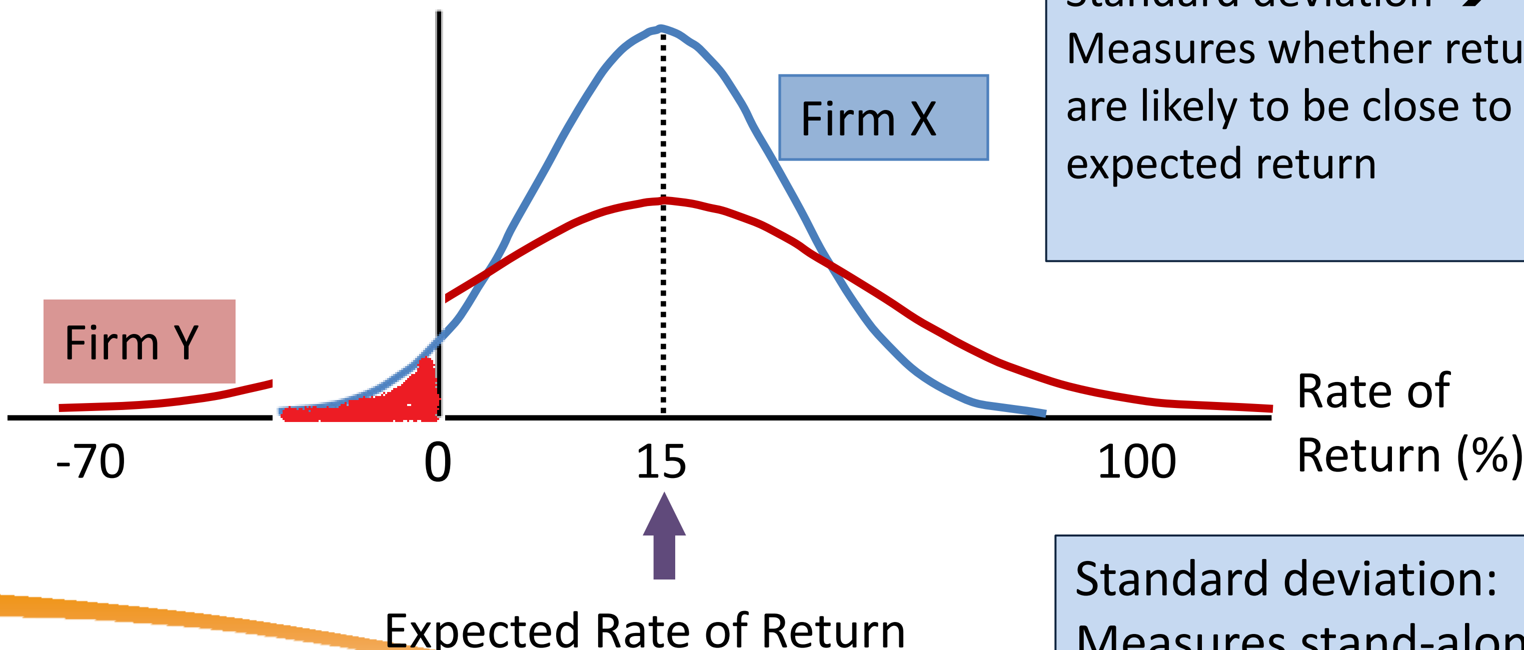
Economy	Prob.	T-Bill	Index	Tech	Defensive
Recession	0.1	5.5%	-17.0%	-27.0%	27.0%
Below avg	0.2	5.5%	-3.0%	-7.0%	13.0%
Average	0.4	5.5%	10.0%	15.0%	0.0%
Above avg	0.2	5.5%	25.0%	30.0%	-11.0%
Boom	0.1	5.5%	38.0%	45.0%	-21.0%

Defensive stocks e.g. Sheng Siong, Walmart while Tech like Tesla and Nvidia are generally riskier

Returns for each state of the world's economy estimated based on future cash flows and current price

Probability Distributions

A listing of all possible outcomes, and the probability of each occurrence.
Can be shown graphically.



Standard deviation → Measures whether returns are likely to be close to the expected return

Standard deviation: Measures stand-alone risk or total risk

What factor drives asset returns?

CAPM says it's the market!

Capital Asset Pricing Model (CAPM)

- Mode linking market risk to asset returns. CAPM suggests that a stock's required return equals the risk-free return plus a risk premium that reflects the stock's risk after diversification.

$$\boxed{r_i = r_{RF} + (r_M - r_{RF})\beta_i}$$

Primary conclusion: The relevant riskiness of a stock is its market risk as measured by beta β_i.

- Single factor here is the market (often the market index)

What is the Market Risk Premium ($r_M - r_{RF}$)?

Average amount of risk ➔ same risk as market portfolio

Meaning

- Additional return over the risk-free rate needed to compensate investors for assuming an average amount of risk.

Size

- Depends on the perceived risk of the stock market and the investors' degree of risk aversion.

Value

- Varies from year to year, but most estimates suggest that it ranges between 4% and 8% per year

Beta β

- Measures a stock's market risk and shows a stock's volatility relative to the market.
  - How sensitive is the stock to market-wide risk factors?

  A stock's beta is the expected change in its return given a 1% change in the return of the market portfolio.

Why market portfolio? Changes in the value of market portfolio are due solely to market-wide events ➔ Market portfolio returns is a good proxy for market-wide events

Comments on Beta

If beta = 1.0, the security is just as risky as the average stock.
If beta > 1.0, the security is riskier than average.
If beta < 1.0, the security is less risky than average.

Most stocks have betas in the range of 0.5 to 1.5.

Extensions of the CAPM

- Single factor not enough (cannot capture all factors that drive stock market).
- Other factors:
  - Size, fundamental ratios, momentum

$$E[R_i] - Rf = \beta m(Rm - Rf) + \beta s(SMB) + \beta v(HML)$$

Factor Pricing Theory

- More general multi-factor framework.
- AI/ML helps by:
  - search for more factors F_1, F_2 etc
  - quantifying the correct relationship: many financial theories construct a linear relationship.Can it be non-linear with no fixed form? Or?

$$E[R_i] \Rightarrow Rf + \beta_1 F_1 + \beta_2 F_2 + \ldots + \beta_k F_k$$

AI/ML has a unique way of specifying relationship correct

AI/ML 'cooking' more factors:

- Search for more factors: alternative data (from Big Data): sentiment factors, image factors (satellite)
- Cooking up (creating) new factors by transforming

Original Data (factors) →  → $\beta_1 AI\_F_1 + \beta_2 AI\_F_2 + \ldots + \beta_k AI\_F_k$

'Transformed' by AI/ML
- Largely unsupervised ML

Some Common Factors (Fundamental)

No	Ratios	Remarks
1	**P/E (Price / Earnings)**	Most common valuation metric; high = growth expectations
2	**P/B (Price / Book)**	Good for banks / asset heavy firms
3	**EV/EBITDA**	Enterprise valuation; universal across industries
4	**EV/Sales**	For unprofitable firms
5	**ROE (Return on Equity)**	Return generated on shareholder capital

- Sometimes, these factors are transformed by unsupervised learning or used as ‘new’ predictors to predict returns.

Challenges in Factor Investing

- **Factor Timing**

Attempting to predict when factors will perform well is extremely difficult. Most evidence suggests staying consistently exposed.

- **Factor Crowding**

As more capital flows into factor strategies, returns may diminish. Popular factors may become overvalued.

- **Implementation Costs**

Transaction costs, market impact, and trading frictions can erode factor returns in practice.

Why factors work?

- **Risk-Based View**

Factors represent compensation for bearing *other* systematic risk. Rational investors demand higher returns for exposures to bad times risk.

- **Behavioral View**

Factors exploit persistent investor biases and market inefficiencies. Examples include overconfidence, anchoring, and herding.

- **Structural View**

Factors arise from market frictions, limits to arbitrage, or institutional constraints that prevent mispricing correction.

Performance Metrics (for portfolio strategies)

- What makes a portfolio 'good'? Aside from profits, there are other important metrics.
- For e.g. have it endured undue risks to generate huge profits? We look at this in other **backtesting results** (we will perform)
- Suppose:
  - you make 4% each year over 5 years buying US treasuries (buy and hold)
  - You make 5% each year over 5 years with BTC (buying and selling)

Which is better?

Performance Metrics

- **Why Multiple Metrics?**

Different metrics capture different aspects of performance: returns, risk, tail risk, downside protection, and consistency. No single metric tells the complete story.

- **Examples of Metrics:**
  - **Return-Based:** Total return, CAGR, Excess return
  - **Risk-Adjusted:** Sharpe, Sortino, Information ratio
  - **Drawdown** Max drawdown, recovery time
  - **Statistical** Volatility, skewness, kurtosis

The Sharpe Ratio

- The most widely used risk-adjusted performance metric

Sharpe Ratio = $(Rp - Rf) / \sigma_p$

- **Rp:** Portfolio return
- **Rf:** Risk-free rate
- **σ_p:** Standard deviation of portfolio returns

Interpretation

Measures excess return per unit of total risk. Higher is better. A Sharpe ratio of 1.0+ is good, 2.0+ is excellent, 3.0+ is exceptional.

Advantages

Simple, widely understood, allows comparison across strategies

Limitations

Assumes normal returns, penalizes upside volatility equally

Sharpe Ratio: Example

Comparing two strategies

Strategy A	Strategy B
Annual Return: 15% Risk-free Rate: 2% Volatility: 10% SR = (15% - 2%) / 10%=1.3	Annual Return: 20% Risk-free Rate: 2% Volatility: 18% SR = (20% - 2%) / 18% **= 1.00**

Strategy A has better risk-adjusted returns despite lower absolute returns.
It generates more return per unit of risk taken.

The Sortinio Ratio

- Focus on downside risk only

  Sortino Ratio = (Rp - Rf) / σ_d

  - **Rp:** Portfolio return
  - **Rf:** Risk-free rate (or target return)
  - **σ_d:** Downside deviation (only negative returns)

Key Difference from Sharpe

Only penalizes downside volatility, not upside. More appropriate when returns are not symmetrical or when investors only care about losses.

When to Use

Preferred for strategies with asymmetric returns, options strategies, or when distinguishing between "good" and "bad" volatility matter.

Maximum Drawdown

- Measuring peak-to-trough decline
  Max DD = (Trough - Peak) / Peak
  Largest percentage loss from a historical peak

What It Shows
The worst loss an investor would have experienced during any period. Critical for understanding pain and risk of ruin.
Example
Portfolio grows from $100k to $150k, then falls to $90k. Max DD = ($90k - $150k) / $150k = -40%
Recovery Time
How long it takes to return to the previous peak. A 50% loss requires 100% gain to recover.

Metrics for different investors

Risk-Averse Investors
Focus on: Max drawdown, Sortino ratio, downside deviation, VaR. Prioritize capital preservation.

Growth Investors
Focus on: CAGR, Sharpe ratio, Calmar ratio. Willing to tolerate volatility for returns.

Active Managers
Focus on: Information ratio, alpha, tracking error. Demonstrate skill vs benchmark.

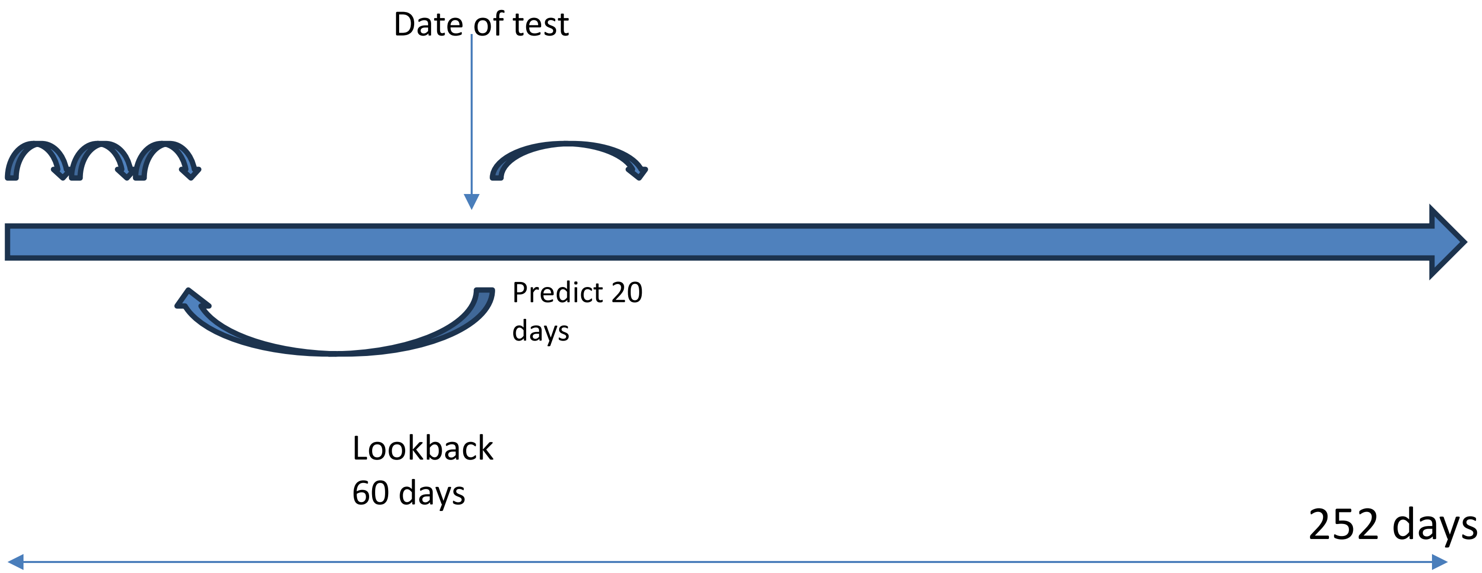
Institutional Investors
Focus on: VaR, CVaR, tail risk metrics, skewness. Manage portfolio-level risk and regulatory requirements.

Testing your strategy with Backtesting

- Backtesting is simulating an investment strategy using historical market data to see how it *would have* performed, revealing its potential profitability, risks, and weaknesses (like drawdowns) before risking real money, helping investors refine asset allocation, optimize rules, and build confidence in a data-driven approach, not just assumptions. (partial AI generated)

- It's like a "time machine" for your investments, allowing you to test different scenarios and market conditions and understand how its performance metrics are.

Illustration of AI/ML Backtesting



- On date of test, use the XGBoost (or other AI/ML) model to find the relationship between the factors and the projected future return known on hindsight (for each stock).
- Sort the predicted return. Long the highest value and short the lowest value at the end of the month.
- At the next month, do the same and re-adjust the portfolio based on the long/short stocks.

Common Pitfalls in Backtesting

1. Relying on Single Metric
No single metric captures all aspects. Always use multiple complementary metrics.

2. Short Time Periods
Metrics can be misleading over short periods. Use at least 1-3 years of data if possible.

3. Ignoring Non-Normality
Sharpe ratio assumes normal returns. Check skewness and kurtosis for tail risk.

4. Survivorship Bias
Failed strategies disappear from databases, inflating reported metrics.

5. Data Snooping
Optimizing for past metrics doesn't guarantee future performance. Choosing the year it works.

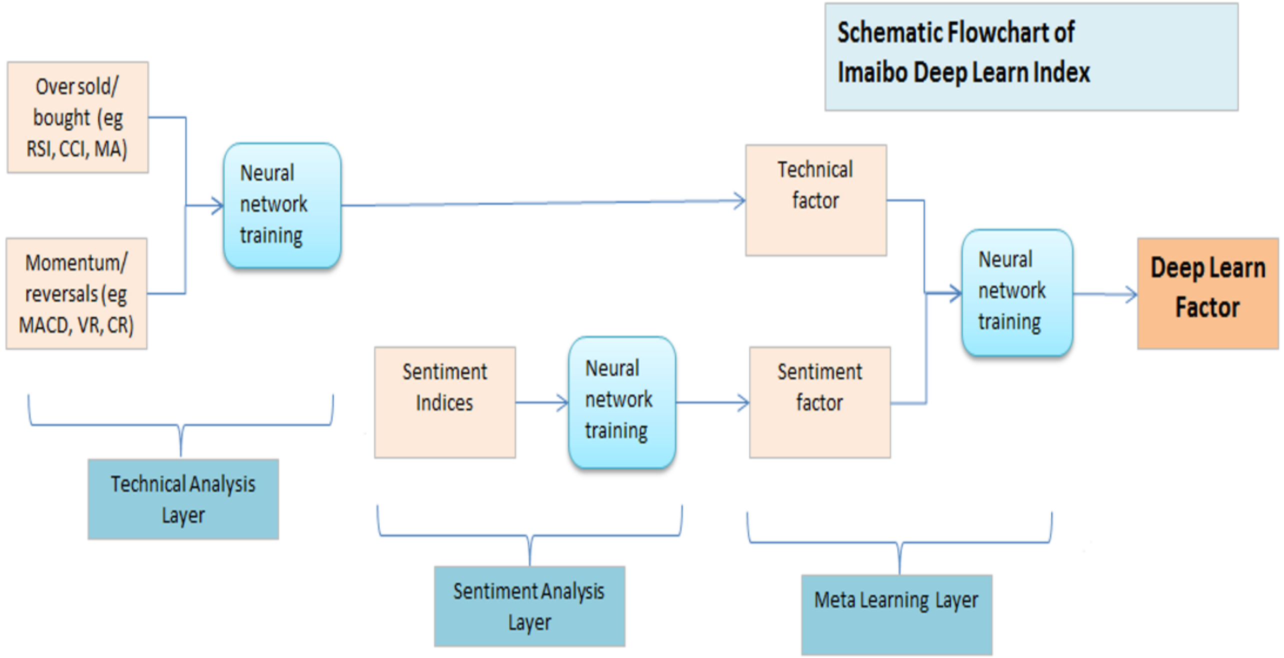
AI/ML Methods

- How have the AI/ML methods worked in backtesting?

- Not possible to cover every AI/ML method used to uncover this relationship. We look at 2 methods (more promising)
  - Ensemble methods: Boosting methods
  - Logistic regression (multinomial)

Why not more 'advanced' methods?

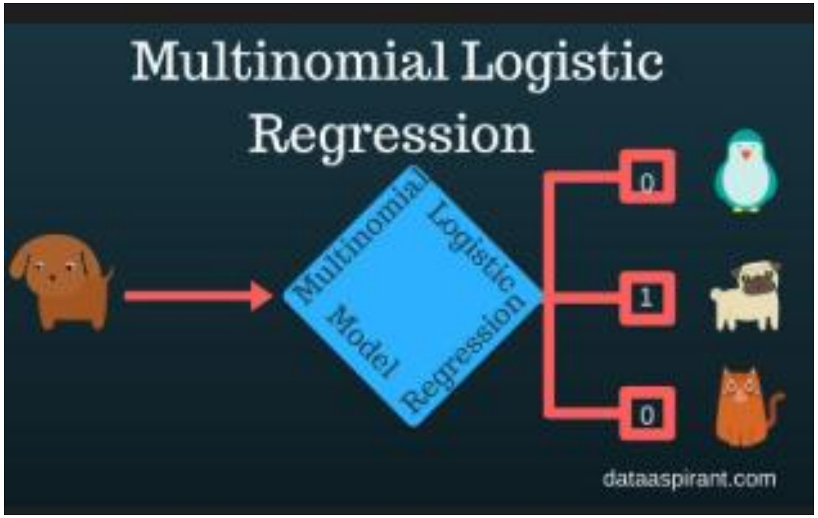
- Eg: deep learning, attention models etc.
  - A good model is judged not by its sophistication but *whether it is profitable* in actual operations.
  - Harder to understand and over-fit in operations



(MultiNomial) Logistic Regression

- Extension of binomial logistic regression. Instead of predictive variable having only 2 classes, there are a multiple of categories (usually n ≥3), modelled separately by n – 1 equations.

- Useful in predicting markets as:
  - Bull market
  - Rangebound
  - Bear market



Ordered Logistic Regression

- Uses proportional odds logit model to accommodate the order of the labels (High > Average> Low) market scenarios matters.
  - Contrast with predicting a bank customer's region (no order as in Singapore, HK, or Japan).

- For ordered categories $\mathrm{Bear}(1) < \mathrm{Rangebound}(2) < \mathrm{Bull}\ (3), \mathrm{model:}$

$P(Y \leq 1)$ and $P(Y \leq 2)$
  - Outputs the odds that the predicted variable is a bear and bear + rangebound?

- Assign predicted variable based on the calculated odds.

Ensemble Models

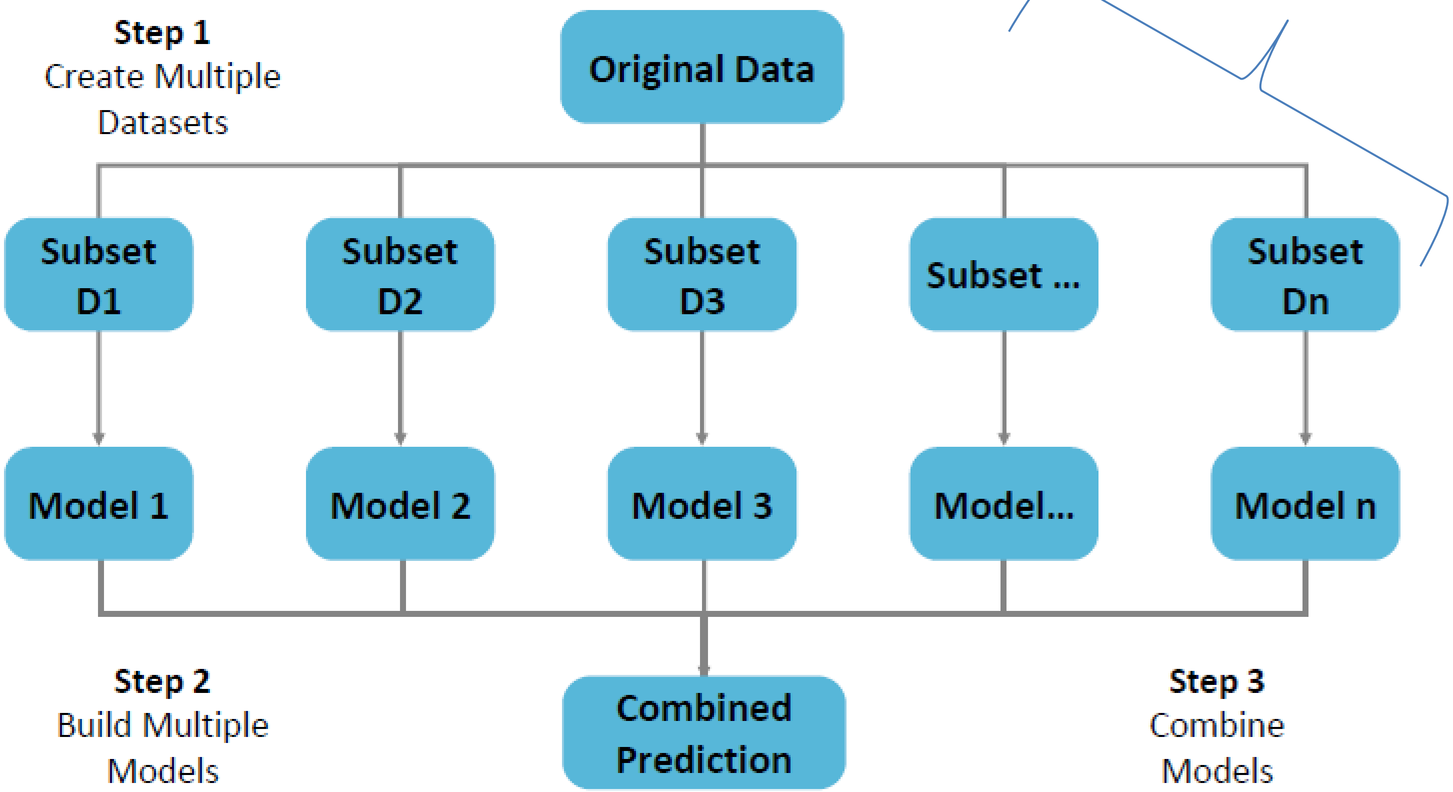
- Aim for low bias and low variance
- Meta Ensemble model**s**
  - Bagging
    - Bagging – wisdom of the collective ‘weak’ learners (usu. homogenous)
  - Boosting
    - Learning sequentially from each other’s weakness
  - Stacking
    - Learning from the best of each other

Bagging

- Parallel set of (weak) learners with final result by voting.

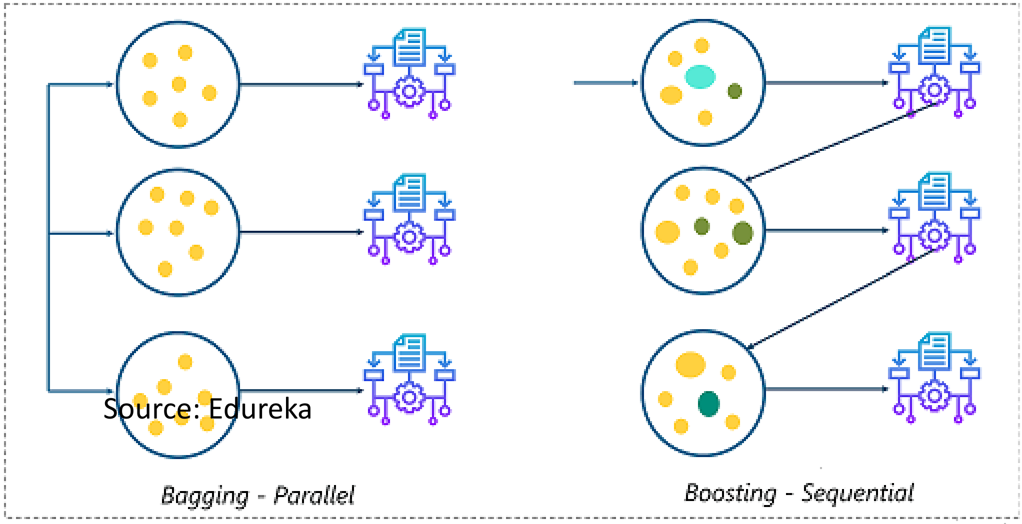
- Useful for high variance data.

Models fed independent datasets by bootstrapping

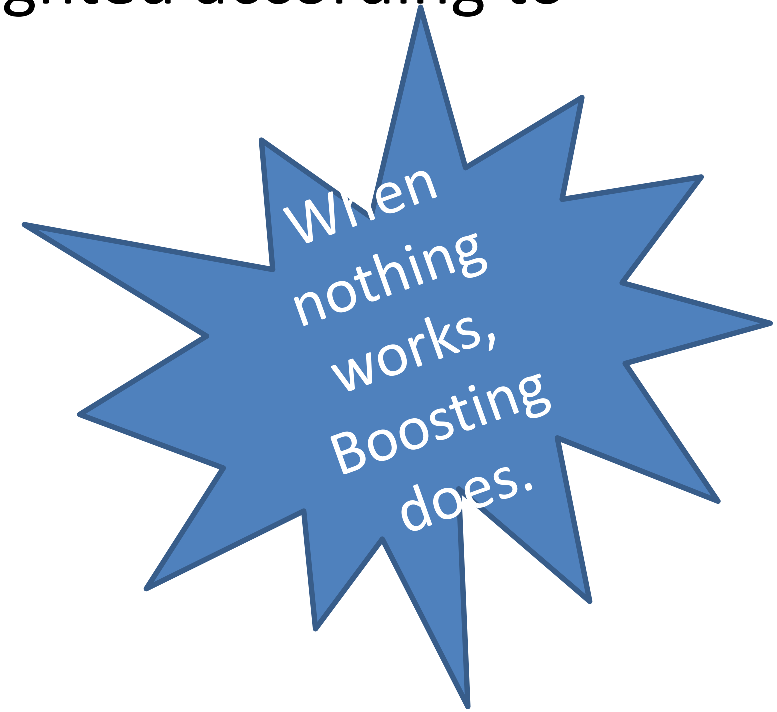


Boosting

- Generally, boosting is a sequential learning model of individual ‘base models’, each weighted according to its accuracy.



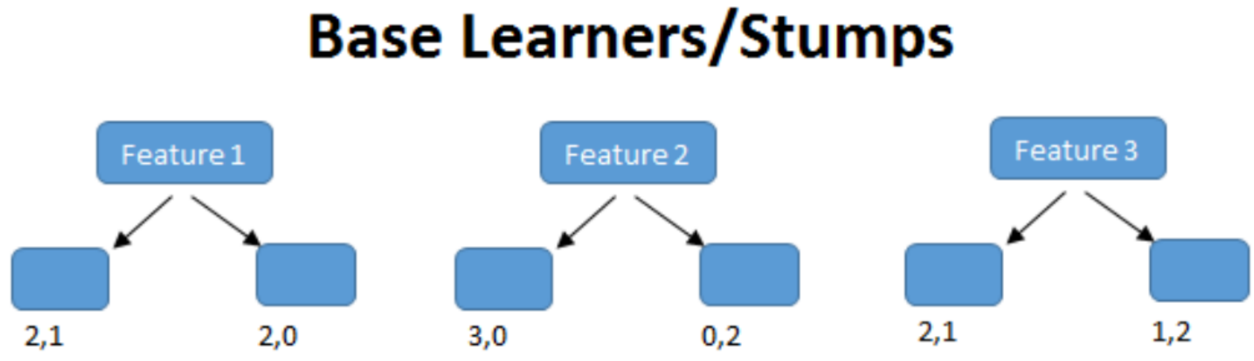
Source: Edureka



ADA Boosting

- Ada Boosting or adaptive boosting trains each base model sequentially, by adjusting its weight – with more weight on ‘more accurate’ models.

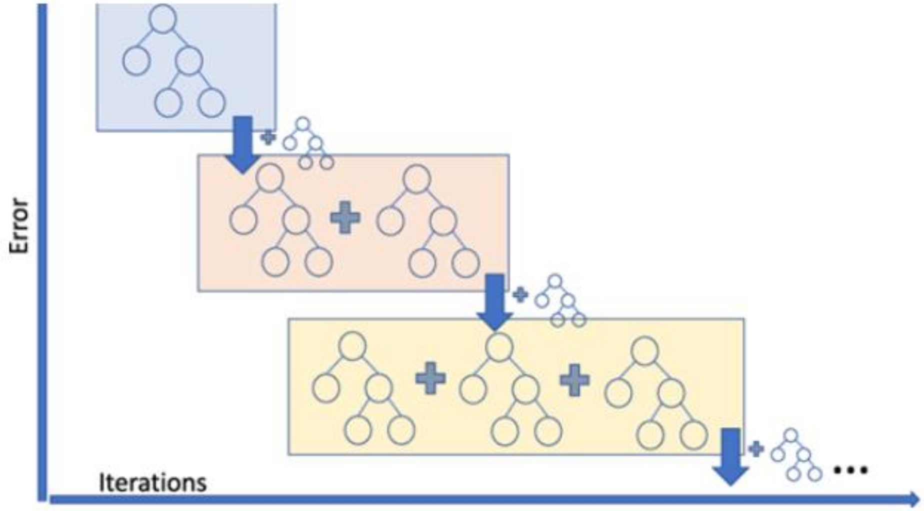
- Each base model is usually a decision stump (decision tree with single level).



- Useful to lower bias and account for non-linear effects.

Gradient Boosting

- Both sequential base learners.
- Gradient boosting has arbitrary loss function, unlike ADA boosting has an exponential loss function.
- ADA looks at the actual error of classification. While gradient boosting focuses on the derivative change on the errors.



Conclusions (Financial Theory)

- Factor pricing models extend CAPM to better explain asset returns through multiple systematic risk factors
- Common factors include market, size, value, momentum, profitability, and investment
- Factors may represent risk compensation, behavioral biases, or market frictions
- Applications include portfolio construction, risk management, and performance evaluation
- Factor investing faces challenges from timing difficulty, crowding, and implementation costs

Conclusions (AI/ML)

- Role of AI / Machine Learning

ML helps discover, transform, and combine factors, handle large alternative datasets, and model non-linear relationships beyond classical theory.

- Core ML Methods Covered
    - Logistic Regression (Binomial / Multinomial / Ordered) for market regime prediction
    - Ensemble Methods: Bagging (variance reduction) & Boosting (bias reduction, non-linearity)
- Model Selection Philosophy

Prefer interpretable, robust models over complex black-box methods; profitability and stability matter more than sophistication.